

NEXORA AI

Launch Readiness Report

Kenya-first Business Operating System

Generated 2026-06-30 · v1.0.0

Executive Summary

NEXORA AI is an all-in-one business operating system designed for Kenyan SMBs and mid-market: POS, inventory, CRM, accounting, payroll, HR, analytics, AI assistant, and chama/group-money primitives. This document covers what shipped in the super-update, what remains before go-live, and the exact production checklist.

Shipped in this session: frontend security hardening, USB+camera barcode scanner, role-based access control with delegated permissions, mobile-first responsive nav, Swahili i18n, eTIMS + M-Pesa Daraja + WhatsApp + Chama integrations, full Node.js + Postgres backend, Capacitor mobile wrap, Tauri desktop wrap.

1. What Happened — Super-Update Scope

1.1 Foundation & Security (P0)

- TypeScript strict mode enforced in build path (was bypassed).
- .env.example sanitized — VITE_* secrets stripped (were leaking into client bundle).
- Store split: useUiStore (theme/sidebar only persisted) + useAuthStore (token in sessionStorage, no role escalation via localStorage) + useNotificationsStore.
- React Query: smart retry (no retry on 4xx), 5-min staleTime, 30-min gcTime.
- Per-page ErrorBoundary inside SuspenseWrapper isolates page crashes.
- PWA registered: offline-ready toast + update toast.
- Security headers on dev server: X-Frame-Options=DENY, Referrer-Policy=strict-origin-when-cross-origin, Permissions-Policy.
- Vite build: manual chunks (react/query/charts/radix/icons/i18n/pdf/scanner) — better cache locality.

1.2 Barcode Scanner (priority)

- src/hooks/useBarcodeScanner.ts — USB / Bluetooth keyboard-wedge detector. Distinguishes scanner bursts (50ms gap) from human typing.
- src/components/scanner/BarcodeScannerModal.tsx — camera scanner using ZXing (EAN-13, UPC-A/E, Code-128, QR). Torch toggle. Manual entry fallback.
- playScanBeep() — WebAudio square-wave beep, success/fail variants.
- POS integration: USB-wedge auto-adds anywhere on page; Scan button opens camera modal; search-bar Enter does lookup; lookup matches barcode OR SKU.
- Receipt numbers now crypto.randomUUID()-based (was Math.random — collision risk).

1.3 RBAC Overhaul + Supervisor Role

Root cause of "cashier sees employee dashboard": both roles were level=1, so the dashboard router collided. Fixed by replacing level-based logic with an explicit role! dashboard map.

- Role union extended: added supervisor (floor oversight) + director (kept ceo as alias).
- PERMISSION_MATRIX: explicit role! path[] list. roleCan(role, path) is now the s
- SupervisorDashboard (new): shift open/close, pending voids/refunds/discount approvals, cashier-on-shift list, hourly sales pace vs target, floor alerts (printer offline, low till, restock).
- CashierDashboard: refocused on selling (open POS button, my sales, payment breakdown, recent transactions) — no longer employee surface.
- window.location.assign removed (SPA-breaking) — replaced with useNavigate.

1.4 Delegated Permissions (capability ACL)

Owner can grant ANY of 60+ capabilities to ANY user without changing their role. Manager who gets users.invite cap can hire staff; cashier who gets pos.refund cap can process returns without becoming supervisor.

- src/constants/permissions.ts: CAPABILITIES list (pos.refund, users.invite, etims.file, mpesa.b2c, chama.disburse, ...) + ROLE_CAPS defaults per role.
- src/utlis/permissions.ts: resolveCaps(user) = defaults + permissionsAdd - permissionsRemove.
- User type extended with permissionsAdd, permissionsRemove, customRoleId.
- CustomRole type for fully bespoke role definitions.
- src/pages/Settings/PermissionsPanel.tsx — UI to grant/revoke per user (toggles, role-default badge, override indicator).
- Backend mirror: src/lib/rbac.ts + PATCH /api/users/:id/grant (gated by users.grant cap).

1.5 Mobile Navigation Rebuilt

- Sidebar: slide-in drawer on <lg (280px) with backdrop, permanent on lg+. Auto-close on route change. Body scroll-locked while open.
- Topbar: hamburger button visible <lg, left snaps to 0 on mobile.
- AppLayout: ml-0 on mobile, lg:ml-[68|260px] desktop.
- New MobileBottomNav (lg:hidden): 5-slot bottom tab bar (Home/POS/Al/Alerts/More) with safe-area inset padding and unread alert badge.
- POS responsive: products full-width on mobile, cart drops below as full-width panel + floating "Cart (n)" pill scrolls user

1.6 i18n — English + Swahili

- i18next + react-i18next + LanguageDetector. I18nGate blocks render until ready.
- Locales: en-KE, sw-KE. Coverage: nav, POS, checkout, common, errors, chama.
- <html lang> updates with current language for assistive tech.

1.7 Service Layer + Kenya Wedge

- ky HTTP client (src/services/api.ts): bearer token from authStore, 401!clear+red
- React Query hooks for all 14 services + mutations (useCreateTransaction, useUpsertCustomer/Product/Employee).
- Offline queue (src/modules/offline/queue.ts): IndexedDB-backed via idb-keyval. Enqueue/pending/flush/online-flusher.
- eTIMS client: fileEtims(txnId) auto-falls-back to queue when offline or API fails.
- M-Pesa client: stkPush + stkStatus typed wrappers.
- WhatsApp share: receipt + product deep links with KE phone normalization (07X).
- Chama page: groups, members, contributions, rota, loans.

1.8 Backend — Node.js + Postgres

- Fastify 5 + Prisma 6 + Postgres 18 + Redis 8 + JWT (Argon2id) + Zod validation + Pino logger + Swagger UI at /docs.
- Multi-tenant schema (every row scoped to workspaceId): Workspace, User, CustomRole, RefreshToken, AuditLog, Branch, Employee, Product, StockLevel, StockMovement, Customer, Transaction, CartItemRecord, Invoice, Lead, Supplier, PurchaseOrder, PayrollRun, Attendance, LeaveRequest, ChamaGroup/Member/Contribution/Loan/RotaSlot, Notification.
- Auth: /register, /login (sets HttpOnly refresh cookie), /refresh, /logout, /me. Argon2id passwords. SHA-256-hashed refresh tokens.
- Capability mirror in src/lib/rbac.ts. requireCap(cap) Fastify preHandler.
- Module endpoints: users (CRUD + grant), products (CRUD + barcode lookup), transactions (POST decrements stock, records movements, fires eTIMS), etims (file + cancel), mpesa (STK + callback), ai (chat + SSE stream to Claude Haiku 4.5).
- Docker compose: Postgres + Redis. Multi-stage Dockerfile (Node 24 alpine).
- CI workflow: lint + test + prisma migrate + build, all gated on Postgres service.
- scripts/boot.sh: one-shot — docker up ! wait pg ! migrate ! seed ! dev.
- prisma/seed.ts: demo workspace + owner@nexora.co.ke / demo1234 + Nairobi HQ branch.

1.9 Mobile (Capacitor)

- capacitor.config.ts — iOS + Android wrappers with SplashScreen, StatusBar, Keyboard, BarcodeScanner (@capacitor/mlkit/barcode-scanning), PushNotifications, LocalNotifications.
- App ID: co.ke.nexora.ai · web dir: dist
- Pipeline: npm run build ! npx cap sync ! npx cap open ios|android ! Xcode/Android

1.10 Desktop (Tauri)

- src-tauri/{tauri.conf.json, Cargo.toml, src/main.rs, build.rs}.
- Targets: Windows (MSI + NSIS), macOS (DMG), Linux (deb + rpm + AppImage).
- Plugins: updater, notification, shell, fs, dialog, os, process.
- CSP: connect-src restricted to nexora.co.ke + safaricom + KRA.
- Auto-updater endpoint stub for releases.nexora.co.ke.

1.11 Polish (Wave 4)

- Sparkline IDs: useId() (fixes shared linearGradient bug when multiple sparklines same color on one page).
- Sparkline + chart wrapper: role="img" + aria-label.
- Per-page ErrorBoundary wrap (was single global).
- Smart React Query retry: no retry on 4xx.

1.12 Intelligence (Wave 5)

- src/utls/pdf.ts: 80mm thermal receipt PDF (with eTIMS QR slot) + A4 report PDF (jsPDF + autotable).

- `src/hooks/useAiStream.ts`: SSE streaming client with abort/cancel.
- Backend POST `/api/ai/chat/stream`: Anthropic stream proxy, model `claude-haiku-4-5-20251001`.

2. What Should Be Added Next

2.1 Page-by-Page React Query Migration

Pages still import mock data directly. Migrate to React Query hooks. Mock service layer already returns the right shapes, so switching is mechanical.

Page	Current direct import	Replace with
POS	products, customers	useProducts(), useCustomers()
Inventory	products, movements	useProducts(), useStockMovements()
Customers	customers	useCustomers()
Employees	employees	useEmployees()
Branches	branches	useBranches()
Suppliers	suppliers	useSuppliers()
CRM	leads	useLeads()
Accounting	invoices, transactions	useInvoices(), useTransactions()
Payroll	payrollRuns, employees	usePayrollRuns(), useEmployees()
HR	leaveRequests, employees	useLeaveRequests(), useEmployees()
Analytics	mock + Math.random	useTransactions() + useMemo aggregation
Reports	invoices, transactions, payroll	corresponding hooks
Dashboard variants	mock	corresponding hooks

2.2 Kenya Wedge UI Wiring

- Checkout success: render eTIMS QR code, receipt number, KRA SDC ID. Use buildReceiptPdf for "Print" button.
- Checkout M-Pesa path: replace mock with stkPush() + countdown polling for callback (60s timeout).
- PWA offline POS: background-sync registration in service worker; reconcile queue on online event.
- WhatsApp share button on receipt and invoice screens.
- Chama: build /chama/[id] group detail page (members, contributions, rota, loans tabs).
- AI Assistant: enrich system prompt with eTIMS deadlines, current month VAT pre-fill from KRA, top alerts.

2.3 UX / Accessibility (WCAG 2.2)

- Swap CommandPalette overlay for @radix-ui/react-dialog (focus trap + role=dialog).
- Tooltip !' @radix-ui/react-tooltip (keyboard-triggerable). Custom Tooltip is hovered
- Chart visually-hidden <table> mirror for screen readers.
- Table rows with onClick: wrap in <button> or add tabIndex={0} + Enter/Space handler.
- Touch-target audit: POS quantity ±, Toast close — minimum 44×44px.
- Memoize all Math.random() calls in component bodies (Analytics retention, KPI sparklines).

2.4 Backend Module Completion

- customers/routes.ts: CRUD beyond stub.
- invoices module: create, send (email + WhatsApp), eTIMS file on send, write-off, payment recording.
- payroll/routes.ts: run (per-month, multi-mobile-money disbursement), approve, P9 PDF export.
- hr/routes.ts: leave requests, attendance, discipline tickets.
- employees/routes.ts: hire, terminate, role + permissions.
- suppliers, branches, leads, purchase-orders: CRUD.
- notifications: WebSocket push, mark-read, push tokens registration.
- chama/routes.ts: group CRUD + contributions + loans + rota generation worker.
- BullMQ workers: etims-filer (retry queue), mpesa-reconciler (balance pull), payroll-runner, reorder-alerts.

2.5 Intelligence Build-out

- CRM Kanban DnD using @dnd-kit/sortable (deps already installed).
- Reorder-point background worker: hourly check, draft PO when stock < reorderPoint.
- Payroll multi-mobile-money disbursement UI (M-Pesa + Airtel Money + T-Kash + bank).
- KRA P9 payslip PDF template.
- Sales/inventory/customer report exports (PDF + CSV).
- AI tool-use: let the assistant query the database via JSON-schema tools (revenue this month, top products, etc).

2.6 Compliance & Operational

- KRA eTIMS production credentials (TIN, BHF ID, API key).
- M-Pesa Daraja production credentials (Consumer Key/Secret/Passkey for the shortcode).
- Privacy policy + Terms of Service published at /legal/.
- Data Protection Act 2019 compliance — DPO contact, breach notification flow.
- Audit log retention policy (7 years for KRA).
- Backup strategy: daily Postgres dump! S3 with 30-day retention.

3. Go-Live Checklist

3.1 Infrastructure

Resource	Spec	Provider
Backend host	2 vCPU / 4GB / 50GB SSD	DigitalOcean App Platform / Hetzner CX22
Postgres	Managed, 4GB RAM, daily backup, PITR 7d	Neon / Supabase / DO Managed
Redis	256MB managed	Upstash / DO Managed
Object storage	S3-compatible, lifecycle 90d glacier	Cloudflare R2 / DO Spaces
CDN + Frontend host	Edge cache, free tier OK	Cloudflare Pages / Vercel
DNS	nexora.co.ke + app.nexora.co.ke + api.nexora.co.ke	Cloudflare
TLS	Let's Encrypt or Cloudflare Origin Cert	—
Container registry	For backend Docker image	GitHub Container Registry
Logs / metrics	Pino !' Loki + Grafana, or Sentry	Grafana Cloud / Sentry
Error tracking	Sentry frontend + backend	Sentry
Uptime monitoring	1-min ping on /health	BetterStack / UptimeRobot

3.2 Secrets to Provision (server-side only)

- JWT_ACCESS_SECRET, JWT_REFRESH_SECRET — generate via: openssl rand -hex 48
- DATABASE_URL — managed Postgres connection string with SSL.
- REDIS_URL — managed Redis.
- KRA_ETIMS_TIN, KRA_ETIMS_BHF_ID, KRA_ETIMS_API_KEY — from KRA OSCU enrollment.
- MPESA_CONSUMER_KEY, MPESA_CONSUMER_SECRET, MPESA_PASSKEY — Daraja portal (production keys, not sandbox).
- MPESA_SHORTCODE — your paybill or till number.
- MPESA_CALLBACK_BASE — public HTTPS URL Safaricom can POST to.
- AI_API_KEY — Anthropic Console.
- S3_ACCESS_KEY, S3_SECRET_KEY, S3_ENDPOINT, S3_BUCKET.
- WHATSAPP_PHONE_ID, WHATSAPP_TOKEN — Meta Business platform.
- SMTP_HOST/PORT/USER/PASS — transactional email provider (Postmark/Resend).
- SENTRY_DSN (server) + VITE_SENTRY_DSN (client, public).

3.3 Pre-Launch Validation

- npm run typecheck — passes (strict TS, zero errors).
- npm run lint — passes.
- npm run test:run — all unit tests green.
- npm run test:e2e — Playwright desktop + mobile flows pass.
- npm run build — production bundle builds, no warnings.
- Lighthouse PWA audit "e 90 across categories."
- Backend npm run build + docker build succeed.
- Prisma migrate deploy is idempotent against a fresh DB.
- Manual test matrix: every role logs in, sees only their dashboard, can only navigate to permitted routes.
- POS end-to-end: USB scanner adds item, camera scans, M-Pesa STK fires, eTIMS receipt printed, WhatsApp share works.
- Offline test: kill network, complete sale, restore network, verify auto-flush of queued txn + eTIMS.

3.4 Store Submission

Channel	Steps
Google Play	1) Create Play Console listing. 2) Generate signed AAB via Android Studio. beta !' production rollout 10% !' 100%. Need: privacy policy URL, store listing (phone + tablet), feature graphic, data safety form.
Apple App Store	1) Apple Developer Program enrollment. 2) Xcode signing + provisioning profile. Connect review (1–3d). Need: privacy nutrition labels, App Store screenshots (6.7" + 5.5"), promo video optional.
Windows	Tauri MSI + NSIS, code-signed via EV/OV cert. Distribute via website (preferred) or Microsoft Store.
macOS	Tauri DMG, signed with Apple Developer ID + notarized via altool/notarytool.
Linux	ApplImage (universal), deb (Ubuntu/Debian), rpm (Fedora/RHEL). Distribute via website + ApplImageHub.

3.5 Launch-Day Runbook

- T-7d: load test the API with k6 (sustained 100 RPS, peak 500 RPS).
- T-3d: dry-run Daraja in production with KSh 1 transaction. Confirm eTIMS filing receipt.
- T-1d: snapshot database. Tag release. Notify pilot customers.
- T - 0: deploy backend !' run prisma migrate deploy !' deploy frontend !' verify /health owner+cashier+accountant.
- T+1h: monitor Sentry, /health uptime, Postgres connections, Redis ops/s.
- T+24h: review audit logs, look for unexpected 403/500s.
- T+7d: gather pilot feedback, ship hotfixes via release channel.

3.6 KPIs to Track Post-Launch

Metric	Target	Source
Activation: % new accounts that complete first sale within 24h	"e 60 %	Mixpanel / PostHog
eTIMS filing success rate	"e 99 %	transactions.etimsStatus
M-Pesa STK success rate	"e 95 %	mpesa callback log
Median POS sale duration (cart !' receipt)	"d 30 s	frontend timing
p95 API latency	"d 250 m s	backend Pino + Prometheus
Crash-free sessions (mobile)	"e 99.5 %	Sentry
DAU/MAU ratio	"e 0.4	analytics
Weekly retention (W4)	"e 60 %	analytics

3.7 Files Produced This Session

Frontend (new):

- src/hooks/useBarcodeScanner.ts + .test.ts
- src/components/scanner/BarcodeScannerModal.tsx
- src/components/navigation/MobileBottomNav.tsx
- src/constants/permissions.ts
- src/utills/permissions.ts
- src/i18n/index.tsx + locales/{en-KE,sw-KE}.json
- src/services/api.ts + queryKeys.ts + queries.ts
- src/modules/{offline,whatsapp,etims,mpesa}/*.ts
- src/pages/Dashboard/SupervisorDashboard.tsx
- src/pages/Chama/index.tsx
- src/pages/Settings/PermissionsPanel.tsx
- src/utills/pdf.ts
- src/hooks/useAiStream.ts
- src/pwa.ts, src/test/setup.ts + utils.tsx

- capacitor.config.ts, src-tauri/* (Tauri scaffold)
- playwright.config.ts, test/e2e/smoke.spec.ts
- .github/workflows/ci.yml, .husky/pre-commit, .prettierrc.json
- HANDOFF.md, NEXORA-LAUNCH-READINESS.pdf (this file)

Frontend (modified):

- tsconfig.app.json (strict on), .env.example (secrets cleansed), package.json (deps + scripts), vite.config.ts (PWA + chunks + headers), index.html (fonts + PWA meta)
- src/App.tsx, src/store/index.ts (split into 3), src/types/index.ts (Role + User extended), src/constants/index.ts (PERMISSION_MATRIX), src/routes/index.tsx (roleCan + per-page boundary + Chama), src/layouts/AppLayout.tsx (responsive), src/components/{ui,charts,navigation/Sidebar,navigation/Topbar}/index.tsx, src/services/index.ts (+ mutations), src/pages/POS/index.tsx (scanner), src/pages/Dashboard/{index,CashierDashboard}.tsx

Backend (created from scratch at /home/davie/WebstormProjects/nexora-backend):

- package.json, tsconfig.json, .env.example, .gitignore, .prettierrc.json, Dockerfile
- prisma/schema.prisma (multi-tenant), prisma/seed.ts
- src/server.ts, src/lib/{env,prisma,logger,auth,rbac}.ts, src/routes/index.ts
- src/modules/{auth,users,products,transactions,etims,mpesa,ai,customers,chama}/routes.ts
- src/modules/{etims,mpesa}/service.ts
- docker/docker-compose.yml, scripts/boot.sh, .github/workflows/ci.yml

Owner login (after seed): owner@nexora.co.ke / demo1234 — change immediately before any production data flows.